Why Jointly Learned Pre-LLM Visual Token Selectors Can Fail: A Diagnostic Case Study

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Abstract. Pre-LLM visual-token compression can save the full visual prefill cost of a vision-language model if a small selector can learn which image tokens the LLM needs for the current question. We test this idea with TC-ViTok, a query-conditioned selector that compresses 576 CLIP patches to k=144 before LLaVA-1.5-7B LoRA training. The result is negative but controlled: the learned selector loses by 4.1/3.4/7.4 points on VQAv2/GQA/POPE to a compute-matched no-compression LoRA-576 reference, and it fails to surpass the strongest question-blind trivial baseline on any of the five evaluated benchmarks. We use this failure as a diagnostic case study rather than proposing TC-ViTok as a compression method. In this stack we identify three failure-mode (FM) hypotheses: coupled selector–adapter oscillation, adaptive-budget collapse, and warm-start dependence. We describe engineering interventions that stabilise training sufficiently for diagnosis: a τ-independent straight-through gradient route, a parameter-free informative token merge that removes a trainable bridge, and a three-stage warm-start curriculum. Seed-42 adaptive-budget continuations identify a non-collapsed region at least through λ_b=0.15, with independent seed-7 replications confirming collapse at λ_b=0.03 and stability at λ_b=0.08, 0.10, and 0.15; λ_b=0.20 shows stronger budget pressure in a single seed. Supporting probes do not recover single-token ablation importance from CLIP/DINOv2 features under the tested linear/MLP probes and sample sizes, while coordinate controls are positive. Evidence is primarily from one LLaVA-1.5-7B stack; a Qwen2-VL-7B portability attempt failed to reach non-trivial task accuracy and is not used as validation.

Keywords: Vision-Language Models · Visual Token Compression · Training Dynamics · Co-Adaptation · Straight-Through Estimator · Query-Conditioned Selection.

1 Introduction

Modern vision-language models (VLMs) such as LLaVA-1.5 [15] attach a large language model (LLM) to a frozen CLIP [19] vision encoder. The encoder maps a 336×336 image to N=576 patch tokens which are prepended to the tokenised

question and fed into the LLM. Because LLM attention cost scales quadratically with sequence length, these visual tokens dominate prefill FLOPs, and the budget is set by image resolution rather than by what the question requires.

A natural design point is a *trainable, query-conditioned pre-LLM* compression module that selects $k \ll N$ tokens before the LLM runs. Pre-LLM placement saves the full prefill cost (unlike mid-LLM pruning [5,22]), and query conditioning lets the budget concentrate on the patches relevant to the current question (unlike query-agnostic approaches [3,14,21]). The intuitive hope is simple: the selector should learn to keep the image patches the LLM needs for the current question. The supervision is, however, highly indirect. The selector receives no token-level labels; gradient arrives only after its selected tokens pass through a frozen projector and a 32-layer LLM whose LoRA [7] adapters are simultaneously changing. The selector therefore trains against a moving downstream model rather than a fixed target, and the LLM adapters in turn train against a moving selector. This is a coupled two-timescale optimisation problem: both parameter groups minimise the same VQA loss, but each changes the gradient field seen by the other.

We built TC-ViTok to test the natural idea above, and found it fails in a strong and informative way. The learned selector loses by 4.1/3.4/7.4 points on VQAv2/GQA/POPE to a matched no-compression LoRA-576 reference, *and* fails to surpass the strongest of three trivial token-selection controls: uniform random drop, uniform grid every 4th patch, and CLS-attention top-k on CLIP's last layer, on every benchmark we measure. The gap to trivial selectors is the phenomenon this paper explains, not a peripheral limitation: if a jointly trained query-conditioned selector cannot beat selectors that do not see the question at all, end-task supervision alone is not reliably discovering token importance in this stack. We therefore use TC-ViTok as a controlled diagnostic substrate rather than as a proposed compression method, and the paper's contribution is the structured account of *why* the gap appears. The negative outcome is not a foregone conclusion: query-conditioned compression modules can succeed when trained with stronger priors or supervision [11,22,12], and one would expect a query-conditioned selector to beat query-blind baselines at the same budget. Our result isolates where that conjecture fails under matched LoRA training and end-task supervision alone.

Contributions.

- **Controlled negative result.** Against the compute-matched LoRA-576 reference and three trivial selectors, the learned query-conditioned selector fails to surpass the strongest question-blind trivial selector on any evaluated benchmark.
- **Single-stack diagnostic case study.** We characterise three failure modes (FM): coupled selector–adapter oscillation, adaptive BudgetPredictor collapse, and warm-start dependence. Each has a mechanism, signature, and reproduction protocol; we do not claim architectural universality.

- **Engineering interventions and diagnostics.** We use a τ-independent straight-through gradient route (Eq. 3), a parameter-free informative token merge, and a three-stage warm-start curriculum. We also report query-agnostic convergence as trajectory-level evidence.
- **Falsifiable reproduction protocols.** We give concrete FM1–FM3 tests for independent stacks, including a distillation-based falsification criterion for FM3.

2 Related Work

Query-agnostic visual token compression. ToMe [3] merges similar tokens via bipartite matching. DynamicViT [20] learns Gumbel-Softmax drop masks distilled from a teacher. EViT [14] ranks tokens by CLS-attention and fuses low-importance ones. PruMerge [21] prunes by CLS rank and merges pruned tokens into nearest neighbours. FastV [5] prunes inside the LLM at layer 2, after visual information has mixed into the text stream. These methods avoid training or rely on strong priors (CLS, layer-2 attention) rather than learn query-aware routing.

Query-aware compression. BLIP-2 [11] uses cross-attention between a fixed bank of learned queries and visual tokens. SparseVLM [22] sparsifies inside the LLM using text-to-visual attention with per-layer recycling. QG-VTC [12] injects the question into the encoder and compresses progressively within it. Focus-LLaVA [23] performs coarse-to-fine instruction-guided selection inside the LLM. No prior work we are aware of combines (a) explicit per-question conditioning, (b) pre-LLM placement on frozen features, and (c) per-sample adaptive budget in the setting studied here; we position TC-ViTok as that combination, primarily so we can study its training dynamics.

Differentiable discrete selection. Gumbel-Softmax [9,16] and straight-through estimators [1] are standard tools for learning subset selectors; AdaViT [17] extends these to per-sample budgets. We adopt Gumbel-Sigmoid for independent binary decisions and use a τ-independent additive gradient route (Section 3.3) to avoid low-temperature gradient starvation in our setting.

Training-dynamics perspective. Unlike inference-only compression, our setting jointly trains a pre-LLM selector with LoRA adapters. This motivates the two-timescale optimisation [4,10] and curriculum-learning [2] lens used later; the compute-matched LoRA-576 control isolates compression from fine-tuning effects.

3 Method

3.1 Problem Setup and the Co-Adaptation Loop

Given image $\mathbf{I}$ and question $\mathbf{Q}$, we learn a policy $\pi_\phi(\mathbf{I}, \mathbf{Q}) \to \mathcal{S} \subseteq \{1, \ldots, N\}$ with $|\mathcal{S}|=k$; the LLM receives only the k selected tokens. Selector parameters ϕ and

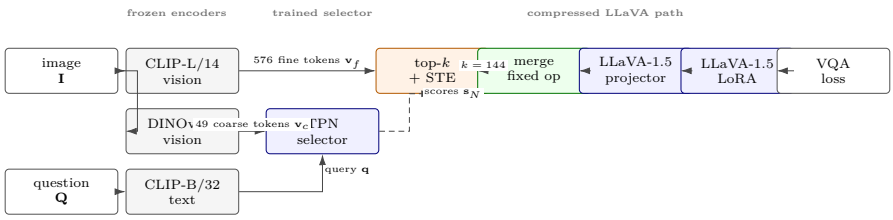


Fig. 1: TC-ViTok dataflow. Frozen encoders provide fine CLIP tokens, coarse DINOv2 features, and the text query. The TPN consumes only $(\mathbf{q}, \mathbf{v}_c)$, scores the 576-token CLIP grid, and the selected tokens pass through fixed merge, the LLaVA-1.5 projector, and LoRA-adapted decoding. Gray nodes are frozen, blue nodes are trained, green is parameter-free, orange is discrete selection, and dashed arrows carry selector scores.

LLM-adapter parameters θ_{LoRA} are jointly trained against the VQA loss:

$$\mathcal{L}_{\text{VQA}}(\phi, \theta_{\text{LoRA}}) = \mathbb{E}_{(I,Q,y)} \left[-\log p_\theta(y \mid \pi_\phi(I, Q)) \right]. \tag{1}$$

This induces a coupled gradient circuit: selector updates shift the LLM input distribution, LoRA updates shift the gradient field the selector sees, and both groups respond to a moving target. The setting is cooperative rather than adversarial: both groups minimise the same VQA loss. We use GAN limit cycles only as an analogy for moving-target optimisation, not as a formal reduction. A closer lens is two-timescale stochastic approximation and bilevel instability [4,10]; we cite these as the appropriate formal lens, not as a convergence guarantee for our nonconvex VLM setting. We observe limit-cycle-like trajectories empirically (Section 4.1), motivating three design choices: parameter-free recovery (no learnable bridge, Section 3.4), warm-freezing LoRA (break the loop in time, Section 3.5), and adaptive-budget training only on a converged base (Section 4.3).

3.2 Architecture

We use three frozen pretrained encoders: DINOv2-S [18] on a 98×98 crop produces $\mathbf{v}_c \in \mathbb{R}^{49\times384}$ that the selector cross-attends; CLIP ViT-B/32 [19] encodes the question into $\mathbf{q} \in \mathbb{R}^{512}$; and CLIP ViT-L/14 [19] produces the LLM input $\mathbf{v}_f \in \mathbb{R}^{576\times1024}$ (aligned to LLaVA-1.5's projector). No gradient flows into any encoder. Thus the TPN consumes $(\mathbf{q}, \mathbf{v}_c)$ and scores the 576 CLIP-L tokens $\mathbf{v}_f$ before selection, merge, and the LLaVA projector.

The *Tokenisation Policy Network* (TPN, ≈887K parameters) cross-attends $\mathbf{q}$ over $\mathbf{v}_c$ with four heads to produce a context vector $\text{ctx} \in \mathbb{R}^{384}$ and head-averaged spatial weights $\boldsymbol{\alpha} \in \mathbb{R}^{49}$. A small score MLP on $[\text{ctx}; \boldsymbol{\alpha}]$ emits 49 coarse scores, bilinearly upsampled to a 24×24 grid and *instance-normalised* to $\mu{=}0, \sigma{=}1$ per sample. Instance normalisation is necessary: without it Adam pushes scores to ±100 within ∼1K steps and Gumbel noise becomes negligible; `tanh` (saturation) and batch norm (single-sample inference) failed analogously.

3.3 Gumbel-Sigmoid STE with τ-Independent Correction

Token selection is a discrete top-k operation. We use Gumbel-Sigmoid STE for independent binary subset selection over 576 positions. With $\mathbf{g} \sim \text{Gumbel}(0, 1)$

and τ annealed linearly $1.0 \to 0.3$, selection is

$$\begin{aligned} \mathbf{n} &= \mathbf{s}_N + \mathbf{g}, \qquad \mathbf{m}_{\text{soft}} = \sigma(\mathbf{n}/\tau), \\ \text{idx} &= \text{topk}(\mathbf{n}, k), \quad \mathbf{m}_{\text{STE}} = \mathbf{m}_{\text{hard}} - \mathbf{m}_{\text{soft}}.\text{detach}() + \mathbf{m}_{\text{soft}}. \end{aligned} \tag{2}$$

The gradient flows through $\mathbf{m}_{\text{soft}}$.

At low τ, $\mathbf{m}_{\text{soft}}$ saturates and $\partial \mathbf{m}_{\text{soft}}/\partial \mathbf{s}_N \approx 0$ for $|\mathbf{s}_N[i]| > 1$, killing the gradient to the selector. Letting $\mathbf{c}$=gather$(\mathbf{s}_N, \text{idx})$, we apply

$$\mathbf{v}_{\text{sel}} \leftarrow \mathbf{v}_{\text{sel}} \odot \big(1 + \mathbf{c} - \mathbf{c}.\text{detach}()\big). \tag{3}$$

Forward: $\mathbf{c}-\mathbf{c}$.detach()$=\mathbf{0}$, so $\mathbf{v}_{\text{sel}}$ is unchanged. Backward: for selected position j, $\partial \mathcal{L}_{\text{VQA}}/\partial \mathbf{s}_N[j] = \sum_d (\partial \mathcal{L}_{\text{VQA}}/\partial \mathbf{v}_{\text{sel}}[j,d])\, \mathbf{v}_{\text{sel}}[j,d]$, a dot product between the VQA gradient and the visual features at j that can remain non-zero even when the sigmoid surrogate saturates. In policy-gradient terms, this is a biased straight-through gradient route rather than an unbiased estimator of the discrete subset objective; we use it as an engineering device to keep selected-token task gradients connected to selector scores. In a matched 50K-step Phase-1 ablation using the same joint-training setup, disabling Eq. 3 worsens final validation next-token-prediction (NTP) loss from 0.3511 to 0.3846 (lower is better), directly validating the correction at the task-loss level. A matched 12K-step Phase-1 joint-training diagnostic run also shows the intended gradient effect: at $\tau \approx 0.10$, the final TPN gradient norm is 0.0324 with Eq. 3 versus 0.0030 without it, while LoRA gradient norms remain comparable (0.4091 vs. 0.4277). A matched no-correction diagnostic with a higher temperature floor ($\tau_{\min}$=0.5) gives a final TPN gradient norm of only 0.0029 and validation NTP loss 0.4069 at 12K steps, indicating that the additive route provides gradient connectivity not recovered by this simple temperature-floor adjustment. The 0.4069 loss is a 12K diagnostic value and is not directly comparable to the 50K no-correction loss; the matched evidence here is the TPN gradient norm.

3.4 Informative Token Merge (Parameter-Free)

Hard dropping of the $N-k$=432 unselected tokens discards their context. We route them into selected tokens via cosine similarity weighted by TPN scores, with normalisation $\tilde{A}_{ij} = w_i A_{ij} / \sum_{i'} w_{i'} A_{i'j}$ where A_{ij} is a row-softmax over cosine similarity at temperature τ_m and w_i is a column-softmax over unselected-token scores:

$$\mathbf{v}_{\text{merged}} = \mathbf{v}_{\text{sel}} + \beta\, \tilde{A}^{\top} \mathbf{v}_{\text{unsel}}, \qquad \beta{=}0.3,\ \tau_m{=}0.1. \tag{4}$$

The merge has *zero new parameters*, eliminates co-adaptation risk with LoRA by construction, and preserves the token count k between training and evaluation. A previous family of eight *trainable* pooler configurations displayed the closed-loop oscillation diagnosed in Section 4.1; the parameter-free merge structurally severs one edge of the gradient circuit and eliminates those failures.

3.5 Adaptive Budget and Training Curriculum

A small ≈25K-parameter *BudgetPredictor* maps selector context features to a per-sample budget fraction $k_{\text{frac}} \in [36/576, 288/576]$. The adaptive variant adds

$$\mathcal{L}_b = \lambda_b \, \mathbb{E}[k_{\text{eff}}/N], \tag{5}$$

to the objective. The collapse modes of this head are characterised in Section 4.2.

All TC-ViTok runs use a three-stage curriculum: **Phase 0** trains the TPN alone (LLM frozen); **Phase 1** closes the joint loop with LoRA warm-frozen for the first 3K steps; **Phase 1 Adaptive** adds the BudgetPredictor on top of a converged Phase 1 checkpoint. We identify this curriculum by elimination: from-scratch joint training and re-orderings of the three stages did not converge in our experiments.

4 A Failure-Mode Case Study for Trainable Pre-LLM Compression

We document three coupled pathologies. Each receives a mechanistic explanation, an empirical signature, and a remedy where one exists. **Scope of evidence.** The controlled evidence comes primarily from one (encoder, LLM, selector) triple: CLIP ViT-L/14 features, LLaVA-1.5-7B with LoRA adapters, and the TPN selector of Section 3.2. We report a failed Qwen2-VL-7B portability attempt for FM1, but do not claim cross-architecture validation. Each pathology is stated with a falsifiable prediction and a reproduction protocol (Section 4.4); whether the predictions hold broadly is an open empirical question.

4.1 FM1: Co-Adaptation Oscillation

Prediction. We predict that other architectures sharing this gradient-circuit structure, a trainable parameter bridge between a learnable pre-LLM selector and adapter-trained LLM weights, will exhibit non-monotonic or limit-cycle-like behaviour in joint training, with cycle period increasing when the selector learning rate is decreased. We state this as a prediction, not a theorem: our evidence is eight selector variants in one (encoder, LLM, selector) triple, and the functional form of the period–LR relationship is not constrained by our two-point sweep (see Signature).

Mechanism. When the selector contains a trainable bridge to the LLM, both groups update at finite rate and each changes the input distribution or gradient field seen by the other. This is not adversarial GAN training: selector and LoRA share the same VQA objective. We therefore frame the GAN analogy as intuition for moving-target optimisation, while the mechanistic claim is narrower: a learnable bridge closes a coupled two-timescale gradient circuit. Removing that bridge removed oscillation in our tested stack.

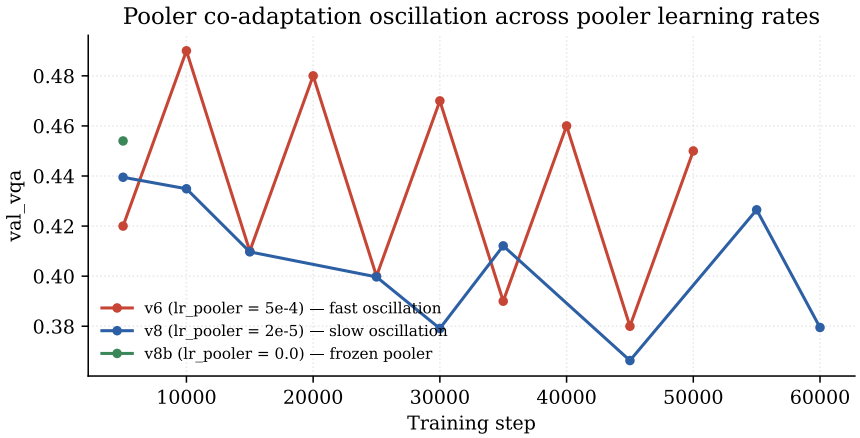


Fig. 2: Representative pooler versions from the FM1 sweep. *Top:* active trainable pooler with $\eta_{\text{sel}}=5\times10^{-4}$, period $\approx$ 5K steps. *Middle:* active trainable pooler with $\eta_{\text{sel}}=2\times10^{-5}$, period $\approx$ 15K steps. *Bottom:* frozen pooler, no oscillation but fixed noise tokens the LLM cannot use.

Signature. Across seven active trainable-pooler configurations we observed oscillatory validation-VQA trajectories. These included cross-attention summarisers with 2–4 learned query slots (varying initialisation and orthogonality constraints), projection bridges, distillation-augmented objectives, and a very-low-LR pooler variant; all kept at least one trainable layer between TPN output and LLM input. Figure 2 shows three representative runs. At $\eta_{\text{sel}}=5\times10^{-4}$ with η_{LoRA} held at 2×10^{-5} (selector 25× faster than LoRA), period $\sim$ 5K steps; at $\eta_{\text{sel}}=2\times10^{-5}$ (both groups matched at the same rate), period $\sim$ 15K steps; freezing the pooler ($\eta_{\text{sel}}=0$) eliminated oscillation but produced fixed noise tokens the LLM could not use. The three-fold period increase under a twenty-five-fold η_{sel} reduction is consistent with the bottleneck shifting from LoRA-limited to jointly-limited dynamics rather than with a simple inverse-η_{sel} relationship. The observed oscillations are consistent with co-adaptation dynamics but do not uniquely identify them; alternative optimisation-path explanations remain possible. We report this as a two-point observation supporting FM1's existence, not a scaling law.

Remedy. Replace the trainable bridge with the parameter-free informative token merge (Section 3.4). With no learnable connector, one edge of the gradient circuit is structurally absent and the loop cannot close. In our LLaVA stack, all active trainable-pooler configurations oscillated; the frozen bridge did not oscillate but was unusable, and the parameter-free TC-ViTok Final does not oscillate.

4.2 FM2: BudgetPredictor Collapse

Prediction. Any adaptive-k head supervised by a monotone-in-k task gradient and the budget penalty in Eq. 5 has a collapse boundary below which $k_{\text{eff}} \to k_{\max}$

Table 1: BudgetPredictor outcome during fine-tuning of the warm-started checkpoint. Collapse: $k_{\text{eff}} > 0.99\, k_{\max}$ for $\geq$ 2K consecutive steps. k_{eff} is the final effective budget; Val is best observed validation NTP loss, not a direct selector-quality ranking because token counts and continuation trajectories differ. Seed-7 replications are completed for $\lambda_b \in \{0.03, 0.08, 0.10, 0.15\}$; "n/a" indicates not replicated in seed 7. Collapse denotes saturation to the full-budget policy: the low Val at seed-7 λ_b=0.03 shows that a collapsed head can still behave as a strong fixed-full-budget model, but it eliminates per-sample token reduction below $k_{\max}$. Stronger budget pressure means non-collapsed k_{eff} below $k_{\max}$ with worse Val than the best stable setting; λ_b=0.20 remains exploratory and single-seed.

λ_b	seed-42 (k_{eff}, Val)	seed-7 (k_{eff}, Val)	Outcome
0.03	287.97, 0.3898	288.00, 0.2466	collapsed in both seeds
0.08	280.15, 0.3368	281.83, 0.3285	stable in both seeds
0.10	271.00, 0.3177	270.37, 0.3391	stable in both seeds
0.15	247.23, 0.3176	245.71, 0.3078	stable in both seeds
0.20	227.85, 0.3291	n/a	stronger budget pressure

uniformly, and an upper region where budget pressure degrades task performance without necessarily causing collapse. The existence of a non-empty stable window is the prediction; its location depends on architecture.

Mechanism. The budget head receives two opposing gradient signals: $\mathcal{L}_{\text{VQA}}$ pulls k_{eff} *up* (more tokens generally ease VQA), $\mathcal{L}_b$ pulls it *down*. When the former dominates, the predictor converges to the trivial saturating policy $k_{\text{frac}} \to k_{\max}/N$ uniformly and the adaptive variant degenerates to fixed-$k_{\max}$ with a dead-weight head.

Signature. Fine-tuning the adaptive variant from a warm-started base, we observe a sharp transition (Table 1). In seed 42, λ_b=0.03 collapses to 287.97/288 within 10K steps and validation loss regresses; $\lambda_b \in \{0.08, 0.10, 0.15\}$ are non-collapsed, and λ_b=0.20 enters a stronger budget-pressure regime. Seed-7 replications reproduce the lower collapse at λ_b=0.03 (final k_{eff}=288.00) and the stable region at $\lambda_b \in \{0.08, 0.10, 0.15\}$ (final $k_{\text{eff}} = 281.83, 270.37, 245.71$). This supports a two-seed replicated non-empty stable adaptive-budget window, while the high budget-pressure boundary at λ_b=0.20 remains single-seed. Interestingly, seed 7 at λ_b=0.03 reaches the lowest validation loss in the table despite saturating at k_{eff}=288: budget collapse is a loss of adaptivity, not necessarily a loss degradation. The low NTP partly reflects its full k=288 budget; because token counts and continuation trajectories differ, Val does not rank selector quality. Adaptive budgeting aims to reduce average token cost below $k_{\max}$; collapse reverts to fixed-$k_{\max}$ and eliminates that saving.

Remedy. Operate inside the measured non-collapsed region, with $\lambda_b \in [0.08, 0.15]$ the two-seed replicated non-collapsed region. Apply the budget head only on a converged base. We expect analogous collapse boundaries in other adaptive-compression systems whenever a budget penalty is the only counterweight to a

Table 2: Warm-start vs. from-scratch outcomes on the joint TPN+LoRA+projector system. Val-VQA is NTP cross-entropy on VQAv2 val (lower is better).

Initialisation	**TPN**	**LoRA** r/α	**Best Val-VQA**	**Steps**
warm-start (P1-Final ckpt)	v1	64 / 128	**0.3177**	40K
from scratch	v1	64 / 128	0.4915 (oscillating)	48K
from scratch	v3	32 / 64	0.4380 $\rightarrow$ 0.39 (regressing)	40K
from scratch	v3	32 / 64†	0.3334 (still descending)	50K

† Image-only TPN; approached the from-scratch plateau near Val-VQA ≈ 0.33 on a single seed by step 50K, not the warm-started ceiling of 0.3177. Rows 1–2 are the controlled warm-start comparison; rows 3–4 vary selector variant and LoRA rank and show the plateau across heterogeneous from-scratch settings.

monotone-in-k task gradient [17,20], but the thresholds in Table 1 are specific to this stack. The λ_b=0.20 row represents a higher-pressure regime that is non-collapsed but has worse Val than the stable region; this upper zone remains exploratory and single-seed.

4.3 FM3: Warm-Start Dependence

Prediction. Joint random-initialisation training of a high-capacity selector and LLM adapters in the tested 7B-scale stack, supervised by end-task loss alone, plateaus above any warm-started equivalent unless an additional supervisory signal (distillation, oracle attention, or a strong prior on the selector) is introduced. This prediction is the weakest of the three because the existence of any one effective supervisory augmentation would falsify it; we invite that test.

Mechanism. A randomly-initialised TPN gives the LLM a near-uniform selection distribution; LoRA faces a flat task landscape and adapts in directions that are information-poor for the converged regime. The selector then receives gradient signal back through a LoRA trained on near-uniform input. Neither parameter group offers the other useful signal until at least one reaches a non-trivial regime. The co-adaptation loop is not oscillating here; it is *stuck*. This places FM3 closer to curriculum and progressive-training failures [2] than to FM1's cyclic dynamics.

Signature. Across four independent large-scale runs spanning two selector variants, three LoRA capacities, and four learning-rate regimes, every from-scratch joint training plateaued well above the warm-started ceiling (Table 2). The only configuration that approached the ceiling started from a converged base.

Remedy. We do not defeat this failure mode. The observed phenomenon is that from-scratch joint training plateaus above the warm-started ceiling under end-task supervision alone. Distillation, oracle importance, or strong selector priors are untested falsification paths rather than completed mitigations in this paper. Thus FM3 is reported as an observed training pathology and falsifiable hypothesis, not as a validated mechanism with a completed mitigation.

4.4 Reproduction Protocols for Other Architectures

Each failure mode carries a reproduction protocol on a different (encoder, LLM, selector) triple, intended to make the predictions in Sections 4.1–4.3 testable by independent groups without re-deriving the analysis.

Reproduction protocols. FM1: instantiate a trainable bridge (learned pooler, cross-attention summariser, or auxiliary projector) between a frozen visual encoder and an adapter-trained LLM, sweep selector learning rates over at least one decade, and test whether oscillation period grows as selector updates shrink; replacing the bridge with a parameter-free recovery operator removed the cycle in our stack. FM2: add an adaptive-budget head with penalty $\lambda_b \, \mathbb{E}[k_{\text{eff}}/N]$, sweep λ_b from 0.03 to 0.30, and test for a collapse boundary below a non-empty stable region. FM3: train from random initialisation with and without an auxiliary token-level distillation signal, such as LoRA-576 single-token-ablation importance or CLS-attention scores as soft TPN targets. FM3 is falsified in this stack if the distillation-augmented run reaches Val-VQA $\leq$ 0.33 within 80K steps, twice the 40K warm-start horizon; without such a counterexample, the observed gap remains a warm-start-dependence diagnosis rather than a solved pathology.

5 Experiments

5.1 Setup

Hardware and data. All experiments run on 2× RTX 6000 Ada (49 GB each) with PyTorch DistributedDataParallel (DDP), BF16 mixed precision, and effective batch size 128. We train on VQAv2 [6] and GQA [8] and evaluate on VQAv2 val2014, GQA val_balanced, POPE [13], MMBench English dev, and TextVQA val.

The missing control. Comparing a LoRA fine-tuned compressive system against an off-the-shelf uncompressed baseline conflates compression with LoRA effects. We introduce LoRA-576: the same Phase-1-Final LoRA adapter and multimodal projector evaluated with all 576 tokens forwarded to the LLM, byte-identical to TC-ViTok Final in rank, alpha, dropout, targets, SFT data, optimiser, and checkpoint step. The architectural comparison is TC-ViTok Final versus LoRA-576, both at ∼60 GPU-h compute; published query-agnostic methods are not compute-matched. This control isolates compression cost from fine-tuning effects, a confound often left open when compression papers compare only to released full-token checkpoints.

5.2 Main Results

TC-ViTok Final costs $-4.1/-3.4/-7.4$ pp on VQAv2/GQA/POPE against LoRA-576 (Table 3). The same pattern persists on an independently retrained

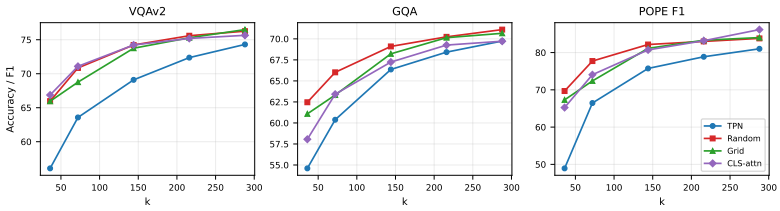


Fig. 3: Token-budget sweep on the Phase 1 Adaptive checkpoint. The learned TPN remains below the strongest trivial selector at every measured k on VQAv2, GQA, and POPE; "strongest" is the per-k maximum of random, grid, and CLS-attention top-k.

seed (Table 4): the gap is not a seed-42 artefact. Against published query-agnostic methods, TC-ViTok underperforms by ∼5 points; those methods use the full publicly released LLaVA-1.5 SFT mix that LoRA-576 also lacks, so the compute-matched architectural cost is the gap to LoRA-576.

Trivial-selector baselines. We evaluate three trivial selectors at the same k=144, same LoRA adapter, and same hard-drop path: uniform random selection, uniform grid (every 4th patch), and CLS-attention top-k on CLIP's last layer (Table 3). *The learned TPN does not surpass the strongest trivial baseline on any of VQAv2, GQA, POPE, MMBench, or TextVQA.* On VQAv2, GQA, and TextVQA the TPN loses to all three trivial selectors. On POPE it marginally exceeds CLS-attention top-k (79.09 vs. 78.94) but falls well short of random (83.92) and grid (82.78). On MMBench it beats random and uniform grid but trails CLS-attention top-k by 0.05 pp (62.0 vs. 62.05), a difference within binomial noise (SE ∼2.5 pp), so we treat the two as statistically indistinguishable. The TextVQA gap is the most striking instance: CLS-attention top-k on the same 24×24 patch grid exceeds the TPN by 10.8 pp, indicating that the joint-training objective did not refine the selector toward text-localisation regions even though that prior is present in CLIP's last-layer attention. The overall pattern is consistent with, but does not independently establish, FM3 (warm-start dependence): the ceiling we approach is bounded by what trivial selection already extracts from this LoRA adapter, and joint end-task supervision through the LLM does not push the learned selector beyond that ceiling. Section 6 gives a compact probe analysis supporting this diagnostic interpretation. For VQAv2 and GQA, the gaps to the strongest trivial selector (2.64 and 3.28 pp) are far larger than the corresponding binomial standard errors (0.16 and 0.22 pp).

Scaling with token budget. The negative finding extends across budgets. On the Phase 1 Adaptive checkpoint, evaluating the same four selectors at $k \in \{36, 72, 144, 216, 288\}$ shows that the learned TPN does not beat the strongest trivial baseline at any measured k on VQAv2, GQA, or POPE (Fig. 3). The gap narrows as k increases but does not reverse, supporting the interpretation that joint end-task supervision through the LLM fails to outperform trivial selection in this regime rather than only at the headline k=144; the failure is not specific to one operating point.

Table 3: Main results on LLaVA-1.5-7B. Headline comparison is TC-ViTok Final vs. LoRA-576. All TC-ViTok rows are single-seed evaluations; binomial standard errors at the listed sample sizes are $\sigma \approx 0.16$ pp for VQAv2 testdev, $\sim$0.22 pp for GQA, $\sim$0.50 pp for the 5K POPE subset, $\sim$2.5 pp for MMBench dev, and $\sim$0.7 pp for TextVQA val. "n/a" means not evaluated.

Method	**Tok**	**VQAv2**	**GQA**	**POPE F1**	**MMB**	**TextVQA**
Published (LLaVA-1.5 SFT, no extra LoRA):						
LLaVA-1.5-7B (full) [15]	576	78.5	61.0	84.4	64.3	58.2
Same-LoRA reference and ours (this work, $\leq$80K steps):						
LoRA-576 (control)	576	76.38	70.55	86.46	61.7	38.4
TC-ViTok v3 (hard drop)	144	72.47	59.87	78.50	n/a	n/a
TC-ViTok Adaptive	36–288	69.43	65.04	75.63	n/a	n/a
TC-ViTok Final	**144**	**72.27**	**67.12**	**79.09**	**62.0**	**25.8**
merge disabled at eval	144	72.27	67.12	79.09	n/a	n/a
zero CLIP query at eval	144	72.67	67.45	79.37	n/a	n/a
Trivial-selector baselines (same LoRA, k=144, hard drop):						
Random drop	144	74.05	70.40	83.92	61.24	29.63
Uniform grid	144	74.65	69.26	82.78	61.93	29.61
CLS-attention top-k	144	74.91	68.16	78.94	62.05	36.57
Published query-agnostic methods (different SFT mix and compute):						
FastV [5]	144	77.1	n/a	n/a	n/a	n/a
LLaVA-PruMerge [21]	144	76.8	n/a	n/a	n/a	n/a
SparseVLM [22]	$\sim$152	78.2	n/a	n/a	n/a	n/a

5.3 Properties of the Converged Selector

Query-agnostic convergence at inference. Zeroing the CLIP query embedding at inference yields $+0.40/+0.33/+0.28$ pp on VQAv2/GQA/POPE on the Final checkpoint (Table 3); Adaptive v2 repeats the sign on two of three benchmarks, with a marginal GQA reversal. The CLIP query channel is not useless during optimisation: image-only and query-conditioned from-scratch runs follow different trajectories in Table 2. At convergence the learned spatial bias suffices and the query becomes redundant. That removing the query improves accuracy indicates that joint optimisation encoded image-distribution priors rather than question-specific routing. We frame this as a trajectory-internal property of training on global-image-distribution data (VQAv2, GQA); independent seeds were not tested.

Token merge: architectural, not benchmark-improving. The merge-disabled ablation (Table 3) is byte-identical to the merged result, and a four-point inference-time β sweep on a 5K subset stays within subset noise. The merge's contribution is *architectural*: it eliminates the trainable bridge that produces FM1 (Section 4.1), not downstream accuracy.

Table 4: Two-seed replication of the matched compression gap. Rows are 5K-sample evaluations from the validation sets at the Phase-1 Final checkpoint (80K steps, identical configuration).

Seed	Model	VQAv2	GQA	POPE F1
42	TC-ViTok Final	72.27	67.12	79.09
42	LoRA-576	76.38	70.55	86.46
7	TC-ViTok Final	72.34	67.58	78.87
7	LoRA-576	76.65	71.36	86.78

6 Discussion

Why TC-ViTok loses to its matched baseline. Three observations explain the $-4.1/-3.4/-7.4$-point gap. First, the selector's gradient signal arrives via a 32-layer LLM and a frozen projector; in contrast to systems that distil from a strong teacher [20], our selector is supervised only by the end-task cross-entropy, a distal signal. Second, the merge β sweep (Section 5.3) shows LoRA absorbs the merge's information flow at every tested β; the LLM adapts to the post-merge representation, shadowing the merge's accuracy contribution. Third, pre-LLM compression cannot exploit the early-layer attention mixing that mid-LLM methods like FastV [5] rely on: information lost before layer 0 is lost for all 32 layers, and the selector must anticipate the LLM's needs without ever seeing an LLM activation. The TextVQA result is not a grid coarseness ceiling: CLS-attention top-k reaches 36.57 at k=144 (Table 3), indicating that end-task training did not move the selector toward CLIP's text-localisation prior.

Supporting probe evidence for the selector ceiling. If learned selection is to improve over trivial baselines by identifying tokens the LLM will causally use, the representations available to the selector should contain recoverable per-token importance. We test this in a matched frozen-teacher probe, using LoRA-576 single-token-ablation loss changes on GQA as a 1000×576 target. Under linear/MLP probes, CLIP ViT-L/14 (R^2=$-0.04/-0.33$) and DINOv2-large [18] ($-0.08/-0.30$) do not recover the target; $R^2<0$ means worse-than-mean prediction. The same pipeline recovers row/column coordinates (R^2=0.39–0.52). A token-permuted CLIP null gives the same negative pattern ($-0.05/-0.31$). At the FastV-relevant L2 depth, visual tokens receive $\sim$89% of attention mass, yet text-to-visual attention is at the mean-prediction baseline ($R^2\approx-0.02$, N=100 pilot smoke test). We report this as a supporting diagnostic result under the tested probes, representations, and sample sizes. Coordinate prediction is a simpler 2D spatial target than 576-way token importance; the positive control validates the pipeline, not the hardness of the target. The result does not rule out weaker signals, richer probes, larger datasets, or alternative causal-importance targets.

Limitations. (1) Primary failure-mode evidence comes from one LLaVA stack; the Qwen2-VL-7B portability attempt did not reach non-trivial task accuracy.

(2) The headline rows are single-seed, although one independently retrained seed checks that the compression gap is not unique to seed 42; we do not claim statistical significance across random initialisations. FM2's lower collapse and stable region are replicated through λ_b=0.15, while the λ_b=0.20 budget-pressure row remains single-seed. (3) The gap to LoRA-576 is not closed, and learned selection never surpasses the strongest trivial selector across our k sweep. (4) The $T_{\text{osc}}(\eta_{\text{sel}})$ relation is a trajectory-level diagnostic, not a scaling law. (5) The probe analysis is scoped to one VLM, one benchmark, two encoder families, linear/MLP probes, N=1000 encoder probes, and an N=100 attention smoke test. (6) FM3's distillation-based falsification criterion is not tested here; the failure mode is characterised but not solved.

7 Conclusion

TC-ViTok was built to test a natural idea: train a query-conditioned selector to keep only the visual tokens the LLM needs for the current question. In one controlled LLaVA-1.5-7B stack, the idea fails informatively: the learned selector loses both to the matched no-compression LoRA-576 reference and to the strongest question-blind trivial selector on every benchmark we measure. We used that failure as a diagnostic substrate and observed three optimisation pathologies in the same stack: co-adaptation oscillation, BudgetPredictor collapse, and warm-start dependence. The τ-independent gradient route, parameter-free informative token merge, and three-stage curriculum are engineering responses that make the diagnostic system trainable; they do not close the accuracy gap. Supporting probes do not recover single-token ablation importance under the tested settings. Evidence is from one stack; the failed Qwen-VL portability attempt is not validation, so the main output is a controlled negative result plus falsifiable hypotheses for independent stacks. In stacks similar to the one studied here, practitioners should first compare against matched CLS-attention or uniform-grid compression baselines before investing in joint pre-LLM selector training under end-task supervision alone.

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